

ROBLIN, MB, Canada

WMO: 715530

Lat: 51.1833N

Lon: 101.3628W

Elev: 540

StdP: 95.01

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-31.0	-36.9	0.1	-33.5	-34.2	0.2	-31.0	12.5	-10.3	11.1	-10.9	2.7	210	0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	28.3	19.9	26.6	18.9	25.0	18.0	21.9	26.4	20.5	24.8	19.3	23.3	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	16.1	24.4	19.0	14.7	22.9	17.7	13.6	21.8	66.7	26.5	61.7	24.7	57.3	23.3	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.0	(b) 9.7	(c) 8.7	(e) -37.7	(f) 31.9	(g) 2.6	(h) 1.7	(i) -39.6	(j) 33.2	(k) -41.1	(l) 34.1	(m) -42.6	(n) 35.1	(o) -44.5	(p) 36.3		
(a) 11.0	(b) 9.7	(c) 8.7	(e) -37.7	(f) 31.9	(g) 2.6	(h) 1.7	(i) -39.6	(j) 33.2	(k) -41.1	(l) 34.1	(m) -42.6	(n) 35.1	(o) -44.5	(p) 36.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-16.3	-14.7	-7.6	2.7	10.0	15.1	17.7	16.6	11.7	3.7	-5.9	-14.0
	DBStd	13.70	8.41	7.75	7.62	6.24	4.78	3.51	3.05	3.39	4.44	5.04	6.79	8.04
	HDD10.0	3800	817	692	545	226	61	4	0	1	31	202	478	744
	HDD18.3	6150	1075	925	803	468	261	108	48	75	203	453	728	1003
	CDD10.0	757	0	0	0	8	60	155	240	204	82	8	0	0
	CDD18.3	64	0	0	0	0	2	9	29	20	4	0	0	0
	CDH23.3	744	0	0	0	2	44	103	291	235	67	2	0	0
	CDH26.7	142	0	0	0	0	8	15	53	54	11	0	0	0
Wind	WSAvg	4.2	4.1	3.9	4.4	4.7	4.9	4.3	3.6	3.6	4.1	4.4	4.2	3.9
Precipitation	PrecAvg	506	22	17	24	29	57	89	77	73	43	35	20	23
	PrecMax	692	42	38	64	80	127	166	166	132	72	99	50	51
	PrecMin	331	4	1	5	3	15	23	24	20	4	11	2	12
	PrecStd	83	10	10	14	19	34	42	32	32	18	19	14	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	2.9	9.9	22.1	27.6	28.5	30.6	31.5	28.2	22.1	10.8	3.5
		MCWB	0.9	1.3	5.6	12.5	15.0	20.1	22.2	20.4	18.0	13.7	7.0	1.6
	2%	DB	-0.3	0.8	6.3	18.0	24.1	26.2	28.0	28.1	25.4	17.4	7.1	0.9
		MCWB	-1.2	0.0	3.9	10.0	13.3	18.0	21.2	19.6	16.8	10.6	4.3	-0.3
	5%	DB	-2.2	-1.3	3.9	15.3	21.6	24.1	26.3	25.9	22.4	14.0	4.5	-1.4
		MCWB	-2.9	-2.0	2.2	8.3	12.5	16.7	20.0	18.6	15.4	9.2	2.5	-2.3
	10%	DB	-4.4	-3.5	2.1	12.4	19.0	22.2	24.8	24.0	19.7	11.2	2.5	-3.4
		MCWB	-4.9	-4.0	1.0	7.1	11.0	15.8	19.2	17.8	13.9	7.6	1.2	-3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.7	6.7	13.0	16.9	22.3	24.6	22.9	20.5	13.9	7.5	1.8
		MCDB	2.1	2.6	8.8	20.1	22.7	26.8	28.5	27.3	25.0	20.7	10.6	3.0
	2%	WB	-1.0	0.0	3.9	10.8	15.0	19.7	22.4	21.2	18.2	11.6	4.7	-0.3
		MCDB	-0.2	0.8	6.0	16.2	21.1	23.3	26.2	26.1	23.3	16.2	6.5	0.9
	5%	WB	-2.9	-2.0	2.3	9.0	13.5	18.2	21.1	19.8	16.3	9.7	2.8	-2.3
		MCDB	-2.4	-1.4	3.8	14.3	20.0	22.3	25.5	24.3	21.1	13.2	4.3	-1.5
	10%	WB	-5.1	-4.2	1.0	7.4	12.1	16.8	19.9	18.6	14.5	8.1	1.2	-4.1
		MCDB	-4.7	-3.6	2.2	12.0	17.7	21.0	23.8	22.8	19.2	11.4	2.5	-3.6
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.8	9.7	10.7	12.1	10.8	11.4	12.1	11.6	9.5	7.8	8.4
		MCDBR	9.3	9.0	9.5	15.0	16.1	13.8	13.8	15.2	15.9	14.8	10.2	9.3
		MCWBR	9.0	8.2	7.4	8.3	7.7	6.9	7.8	7.7	8.4	8.9	7.7	8.6
	5% WB	MCDBR	9.5	8.7	9.2	13.8	13.7	11.7	12.8	13.0	14.3	13.7	9.7	9.4
		MCWBR	9.0	7.9	7.3	8.1	7.4	6.9	7.9	7.6	8.2	8.5	7.4	8.6
Clear-Sky Solar Irradiance	taub		0.246	0.238	0.310	0.323	0.334	0.358	0.369	0.372	0.340	0.280	0.252	0.229
	taud		2.283	2.342	2.239	2.413	2.444	2.398	2.389	2.364	2.442	2.578	2.478	2.358
	Ebn at Noon		792	918	886	920	921	897	880	857	850	861	798	769
	Edh at Noon		64	80	109	104	106	112	112	109	90	64	53	50
All-Sky Solar Radiation	RadAvg		1.27	2.44	3.88	5.07	5.67	5.79	6.17	5.03	3.69	2.13	1.14	0.90
	RadStd		0.09	0.17	0.26	0.36	0.49	0.43	0.29	0.42	0.33	0.29	0.15	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-0.61	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.50	N/A	N/A	N/A	N/A

Nomenclature: See separate page